

Burgers Equation

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1 The Cole-Hopf Transformation

The Cole-Hopf transform is a powerful analytical *linearization* technique. The algebraic structure of some nonlinear partial differential equations allow them to be linearized by a transformation of the solution. That is to say, given certain nonlinear systems, it can be shown that a certain transformation of the unknown solution must solve a different, linear, system. This means that for such a nonlinear equation, we need only solve a linear equation. For illustration, we use $\mathcal{D}$ to denote the nonlinear differential operator, and $\mathcal{L}$ to denote the linear differential operator.

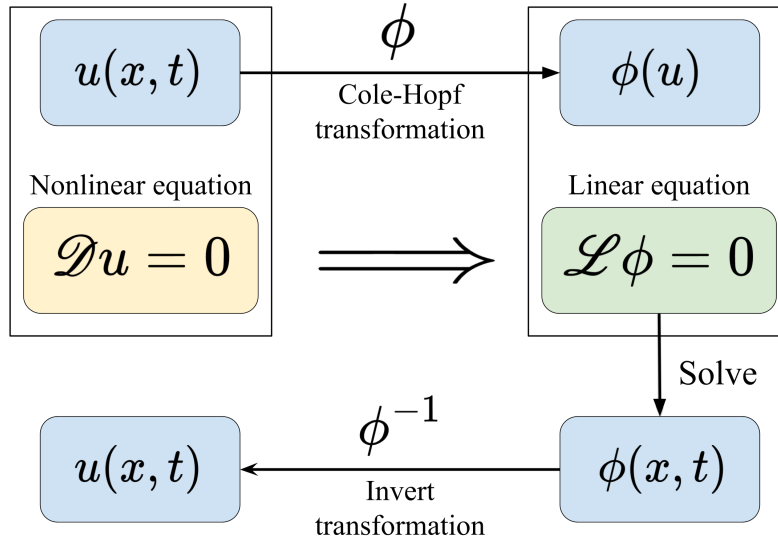


Figure 1: Diagram of the analytical process for solving a nonlinear equation via the Cole-Hopf transformation.

This is a remarkable method that makes it possible to side-step the nonlinearity of certain partial differential equations. For different equations you may need a different Cole-Hopf transformation, and, as one may imagine, the individual transformation must be somehow informative of the nonlinear term in that equation. After all, we pick the transformation so that its algebraic structure untangles this nonlinearity. So of course, the cost of the Cole-Hopf method is that the complexity that it takes away from the equation, it adds to the expression of the solution instead. Indeed, we will see that solutions to simple nonlinear equations are already difficult to handle analytically.

2 Nonlinear Test Equation

Consider the equation

$$u_t - \nu u_{xx} + u^2 = 0, \quad \nu > 0, \quad u(x, 0) = g(x)$$

with 2π periodic boundary conditions.

The first big difference we notice when comparing to the heat equation is that this equation is conceptually much more difficult to solve. In order to solve this PDE we will use the Cole-Hopf transformation to linearize it.

2.1 Applying a Cole-Hopf Transformation

Pick the Cole-Hopf transformation $\phi(u) = e^{-u/\nu}$, and observe what must follow. First, we can say

$$\begin{aligned}\phi_t &= -\frac{1}{\nu} u_t e^{-u/\nu}, \\ \phi_x &= -\frac{1}{\nu} u_x e^{-u/\nu}, \\ \phi_{xx} &= -\frac{1}{\nu} u_{xx} e^{-u/\nu} + \frac{1}{\nu^2} (u_x)^2 e^{-u/\nu}.\end{aligned}$$

Our Cole-Hopf method tells us that if we are dealing with the right kind of equation, and if we have done a good job picking the transformation, then ϕ must solve a partial differential equation that is linear.

The claim is that ϕ actually solves the **heat equation**. With nothing but algebraic substitution and rearrangement, we examine:

$$\begin{aligned}\phi_t - \nu \phi_{xx} &= -\frac{1}{\nu} u_t e^{-u/\nu} - \nu \left(-\frac{1}{\nu} u_{xx} e^{-u/\nu} + \frac{1}{\nu^2} (u_x)^2 e^{-u/\nu} \right) \\ &= e^{-u/\nu} \left(\frac{1}{\nu} u_t + \nu \left(-\frac{1}{\nu} u_{xx} + \frac{1}{\nu^2} (u_x)^2 \right) \right) \\ &= \frac{1}{\nu} e^{-u/\nu} \left(u_t + \nu \left(-u_{xx} + \frac{1}{\nu} (u_x)^2 \right) \right) = \frac{1}{\nu} e^{-u/\nu} \underbrace{\left(u_t - \nu u_{xx} + (u_x)^2 \right)}_{\text{homogeneously satisfied}} = 0.\end{aligned}$$

Where in the last step we remember u annihilates the parenthesized expression by assumption.

So we indeed find it is true that ϕ satisfies the heat equation. Any worries about boundary conditions and initial conditions are also resolved immediately:

(BC): $\phi(x + 2\pi, t) = e^{-u(x+2\pi, t)/\nu} = e^{-u(x, t)/\nu} = \phi(x, t)$, so our transformation is naturally also periodic;

(IC): $\phi(x, 0) = e^{-u(x, 0)/\nu} = e^{-g(x)/\nu}$, we have an explicit expression for the corresponding initial condition.

So let us solve the heat equation for $\phi(x, t)$. We express our solution as

$$\phi(x, t) = \sum_{k=-\infty}^{\infty} \hat{\phi}_k(t) e^{ikx},$$

which gives us

$$\phi_t = \sum_{k=-\infty}^{\infty} \frac{d\hat{\phi}_k}{dt} e^{ikx}, \quad \phi_{xx} = \sum_{k=-\infty}^{\infty} -k^2 \hat{\phi}_k(t) e^{ikx}.$$

Knowing, $\phi_t - \nu \phi_{xx} = 0$, we have

$$\sum_{k=-\infty}^{\infty} \frac{d\hat{\phi}_k}{dt} e^{ikx} = -\nu \sum_{k=-\infty}^{\infty} k^2 \hat{\phi}_k(t) e^{ikx}.$$

Due to orthogonality we equate coefficients, i.e. $\frac{d\hat{\phi}_k}{dt} = -\nu k^2 \hat{\phi}_k$, so $\hat{\phi}_k(t) = \hat{\phi}_k(0) e^{-\nu k^2 t}$. Hence our transformed solution is

$$\phi(x, t) = \sum_{k=-\infty}^{\infty} \hat{\phi}_k(0) e^{-\nu k^2 t} e^{ikx}.$$

We obtain $\hat{\phi}_k(0)$ simply with

$$\hat{\phi}_k(0) = \frac{1}{2\pi} \int_0^{2\pi} \phi(0, x) e^{-ikx} dx = \frac{1}{2\pi} \int_0^{2\pi} e^{-g(x)/\nu} e^{-ikx} dx.$$

Using the reverting transformation $u = -\nu \ln \phi$, we obtain our solution:

$$u(x, t) = -\nu \ln \left(\sum_{k=-\infty}^{\infty} \hat{\phi}_k(0) e^{-\nu k^2 t} e^{ikx} \right).$$

Miraculously, we have arrived at an exact expression for the solution to this test equation.

3 Burgers Equation

Now let's apply our ideas to Burger's equation, whose only difference is an altered nonlinear term:

$$u_t - \nu u_{xx} + uu_x = 0, \quad \nu > 0$$

Burgers' equation can be used to model simple fluid systems, as well as traffic flow. This equation is useful to us, as the nonlinear term of Burgers' equation is the same advective nonlinear term as in the Navier-Stokes equation. So studying it may give us some insight into the important and difficult nonlinear dynamics that we see in fluid flow.

3.1 Applying another Cole-Hopf Transformation

Use the Cole-Hopf transformation $u = -2\nu \frac{\psi_x}{\psi}$. So $u = -2\nu \frac{\partial}{\partial x} \ln \psi$. So we see:

$$\begin{aligned} u_t - \nu u_{xx} + uu_x = 0 &\Rightarrow \nu \frac{\partial}{\partial x} \left(\frac{\psi_t}{\psi} \right) - 2\nu^2 \left(\frac{\partial}{\partial x} \ln \psi \right) \left(\frac{\partial^2}{\partial x^2} \ln \psi \right) - \nu^2 \frac{\partial^3}{\partial x^3} \ln \psi = 0 \\ &\Rightarrow \frac{\partial}{\partial x} \left(\nu \frac{\psi_t}{\psi} - \nu^2 \left(\frac{\psi_x}{\psi} \right)^2 - \nu^2 \frac{\psi_{xx}\psi - \psi_x^2}{\psi^2} \right) = 0 \\ &\Rightarrow \frac{\partial}{\partial x} \left(\frac{1}{\psi} (\psi_t - \nu \psi_{xx}) \right) = 0 \end{aligned}$$

Integration with respect to x yields

$$\psi_t - \nu \psi_{xx} = \psi h'(t),$$

where $h : \mathbb{R}_0^+ \rightarrow \mathbb{R}$ is some arbitrary scalar function of time.

Now we simply apply a transformation $\psi \rightarrow \psi e^h$. Note that this does not change u as

$$-2\nu \frac{\partial}{\partial x} \ln(\psi e^h) = -2\nu \frac{\partial}{\partial x} \ln \psi = u.$$

So we proceed. The transformation turned our equation into

$$\psi_t e^h + \psi h' e^h - \nu \psi_{xx} e^h - \psi e^h h' = 0$$

$$\Rightarrow \psi_t - \nu \psi_{xx} = 0.$$

We see that we have thus reduced Burgers' equation to the heat equation as well.

We know how to take it from here – once again having maintained our periodic boundary conditions, we have

$$\psi(x, t) = \sum_{k=-\infty}^{\infty} \hat{\psi}_k(t) e^{ikx},$$

with which we state

$$\sum_{k=-\infty}^{\infty} \frac{d\hat{\psi}_k}{dt} e^{ikx} + \nu \sum_{k=-\infty}^{\infty} k^2 \hat{\psi}_k(t) e^{ikx} = \sum_{k=-\infty}^{\infty} \left(\frac{d\hat{\psi}_k}{dt} + \nu k^2 \hat{\psi}_k(t) \right) e^{ikx} = 0.$$

Orthogonality gives us

$$\frac{d\hat{\psi}_k}{dt} = -\nu k^2 \hat{\psi}_k,$$

a family of ODEs with solutions

$$\hat{\psi}_k(t) = \hat{\psi}_k(0)e^{-\nu k^2 t},$$

hence giving us

$$\psi(x, t) = \sum_{k=-\infty}^{\infty} \hat{\psi}_k(0)e^{-\nu k^2 t} e^{ikx}$$

and

$$\psi_x = \sum_{k=-\infty}^{\infty} ik\hat{\psi}_k(0)e^{-\nu k^2 t} e^{ikx}.$$

Since $u = -2\nu \frac{\psi_x}{\psi}$, we conclude

$$u(x, t) = -2\nu \frac{\sum_{k=-\infty}^{\infty} ik\hat{\psi}_k(0)e^{-\nu k^2 t} e^{ikx}}{\sum_{k=-\infty}^{\infty} \hat{\psi}_k(0)e^{-\nu k^2 t} e^{ikx}}.$$

So once again, by applying the Cole-Hopf transformation, we successfully reduced Burgers' equation to the heat equation, which allowed us to solve for $\psi(x, t)$ using Fourier series. Then we reconstructed $u(x, t)$ by reverting the transformation, blessing ourselves with an explicit solution.